OPT 1

P1) $q = \begin{bmatrix} x_c \\ \ell \\ \theta \end{bmatrix}$ $\begin{array}{l} \rightarrow \text{pos of cart} \\ \rightarrow \text{length of rope} \\ \rightarrow \text{angle of load} \end{array}$

$c\theta = \cos(\theta)$
$s\theta = \sin(\theta)$

$$L = T - V$$

$$T_{cart} = \frac{1}{2} m_c \dot{x}_c^2$$

$$T_{Load} = \frac{1}{2} m_L \left(\dot{x}_c^2 + \dot{\ell}^2 + (\ell \dot{\theta})^2 + 2 \dot{x}_c \dot{\ell}\, c\theta \right.$$

$$T_{rot} = \frac{1}{2} J_M \omega_m^2 = \frac{1}{2} J_m \left(\frac{\dot{\ell}^2}{r_m} \right)$$

$$V_{Load} = m_L g \ell c\theta$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i \longrightarrow \text{MATLAB}$$

P2) $\dot{x} = f(x,u)$

$$x = \begin{bmatrix} x_w \\ \dot{x}_w \\ \ell \\ \dot{\ell} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}, \quad u = \begin{bmatrix} a_x \\ a_\ell \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\ddot{x}_c = a_x$$
$$\ddot{\ell} = a_\ell$$
$$\ddot{\theta} = \frac{1}{\ell}\left(-g s\theta - a_x c\theta - 2\dot{\ell}\dot{\theta}\right)$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} x_2 \\ a_x \\ x_4 \\ a_\ell \\ x_6 \\ \frac{1}{x_3}\left(-g\, s(x_5) - a_x c(x_5) - 2x_4 x_6\right) \end{bmatrix}$$

$$f(x,u) = \begin{bmatrix} x_2 \\ u_1 \\ x_4 \\ u_2 \\ x_6 \\ \frac{1}{x_3}\left(-g\, s(x_5) - u_1 c(x_5) - 2x_4 x_6\right) \end{bmatrix}$$

One method to compensate for friction effects is by adding friction feedforward compensation. This involves modeling the friction force and incorporating it to the control input

P3)
P5) $r_L = \begin{bmatrix} x_L \\ y_L \end{bmatrix} = \begin{bmatrix} x_c + l\,s(\theta) \\ l_{y_0} - y_L \end{bmatrix} = \begin{bmatrix} x_c + l\,s(\theta) \\ l_{y_0} - l\,c(\theta) \end{bmatrix}$

$r_{L,0} \longrightarrow r_{L,T}$

$l_{y_0} = 0.72$

Optimization Problem.

$$\min_u J(u) = \frac{1}{2}\Delta x(T)^T S \Delta x(T) + \int_0^T \left(\frac{1}{2}\Delta x^T Q \Delta x + \frac{1}{2} u^T R u\right) d\tau$$

$\Delta x = x - x_f$

$S \to$ terminal state, $Q \to$ state deviation, $R \to$ input

$\dot{x} = f(x, u) \sim (P2)$

$x(0) = x_0$

$u_1 = a_x \in [-9, 9]$, $u_2 = a_l \in [-3.5, 3.5]$

$S = Q = \mathrm{diag}([1,1,1,1,1,1])$

$R = \mathrm{diag}([0.01, 0.01])$

Time horizon

$x_c \in [-0.6, 0.6]$

$l \in [0.25, 0.8]$

P4) $L = J(u) + \int_0^T \lambda^T (f(x,u) - \dot{x})\, d\tau$

$L = \frac{1}{2}\Delta x(T)^T S \Delta x(T) + \int_0^T \left(\frac{1}{2}\Delta x^T Q \Delta x + \frac{1}{2}u^T R u + \lambda^T (f(x,u) - \dot{x})\right) d\tau$

First order cds:

1) $\dot{x} = f(x,u)$ with $x(0) = x_0$

2) $\dot{\lambda} = -\frac{\partial H}{\partial x}$ → Hamiltonian, $H = \frac{1}{2}\Delta x^T Q \Delta x + \frac{1}{2}u^T R u + \lambda^T f(x,u)$

3) $\frac{\partial H}{\partial u} = 0 \rightarrow u = -R^{-1} \frac{\partial f(x,u)}{\partial u}^T \lambda$

4) $\lambda(T) = S\Delta x(T)$, $x(0) = x_0$

Numerical soln using GD methods

1) Discretize: - discretize into N intervals with step size $\Delta t = \frac{T}{N}$
- represent states x_k, controls u_k, and multipliers λ_k at each time step k

2) Initialization: - initialize u_k
- compute state trajectory x_k using fwd integration

3) λ Equation: Solve the eqn using terminal cdn: $\lambda_N = S\Delta x_N$

4) Update control input: - Compute the gradient of the Hamiltonian wrt u_k:

$$D_u J = \frac{\partial H}{\partial u_k} = Ru_k + \frac{\partial f(x_k, u_k)}{\partial u_k}^T \lambda_k$$

- Update u_k using step size $\alpha > 0$

$$u_k^{new} = u_k - \alpha D_u J$$

5) $u_k = \max(u_{min}, \min(u_k, u_{max}))$

6) Repeat steps 3-5 until convergence, the change in cost $J(u)$ is below a threshold.